PRACTICAL GUIDE · STEP BY STEP

# Installing Python and JupyterLab

*Two installs, fifteen minutes. No prior Python knowledge assumed — no familiarity with the terminal, no jargon.*

**Everything else installs itself.**

1. **Python**
   the engine that runs the code
2. **JupyterLab**
   the window where you open notebooks

*Commands tested on a real machine · October 2026*

AGENDA

# The path in five steps



## 01

### Understand

The two pieces to install, and three quick checks first



## 02

### Check

Python may already be there: one command tells you



## 03

### Install Python

Windows, macOS or Linux, then a version check



## 04

### JupyterLab

One command to install it, another to launch it



## 05

### Troubleshoot

The five most common problems and their fixes



Only one ordering rule: **Python before JupyterLab**, since JupyterLab is installed through Python.

# Two pieces, each with its own job

THE ENGINE

## Python

It runs the code. It never shows itself: it works behind the scenes.

**Version 3.10 minimum · 3.12.0 tested**

*then*

THE WINDOW

## JupyterLab

It opens notebooks and shows results next to the code. It needs Python to work.

**Installed by Python itself, in one command**

**And that's it.** The thirty libraries the project needs — pandas, plotly and the rest — **install themselves** when the first notebook is opened.

*You don't need to know that list, or type it.*

# Three things to check, one minute

## 01



### Which system?

Windows, macOS or Linux. Commands differ slightly, and every slide gives all three.

## 02



### Can you install software?

On a work computer, installing software is sometimes reserved for IT. Check this **before you start**: it is the only obstacle you cannot remove yourself.

## 03



### An Internet connection

The download is about thirty megabytes for Python, a little more for JupyterLab.

*None of these three points requires any knowledge of Python.*

# Python may already be installed

Open a terminal and type this command. It changes nothing: it simply asks Python to introduce itself.

**WINDOWS — COMMAND PROMPT OR POWERSHELL**

```
python --version
```

*if nothing happens:* `py --version`

**MACOS AND LINUX — TERMINAL**

```
python3 --version
```

*the 3 matters: plain "python" often means version 2*

| WHAT YOU SEE | WHAT IT MEANS | WHAT YOU DO |
|---|---|---|
| **Python 3.10, 3.11, 3.12 or 3.13** | Everything is ready | Skip straight to step 3 |
| **Python 3.9.x or lower** | Too old for the project | Install a recent version — step 2 |
| **"is not recognized"** | Python is not installed, or not registered with the system | Step 2 |
| **The Microsoft Store opens** | It's a Windows shortcut, not Python | Step 2, then see "Troubleshoot" |

*Recorded on a real machine: the expected answer is "Python 3.12.0"*

# Windows: install Python in five moves

1. **Open the download page**
   `python.org/downloads` in your browser.

2. **Download**
   The yellow button offers the Windows version. Any version 3.10 or later will do.

3. **Run the downloaded file**
   It is called `python-3.12.x-amd64.exe`.

4. **TICK the box at the bottom of the window**
   "Add python.exe to PATH". This is the decisive move of the whole install.

5. **Click "Install Now"**
   Then close the window when it shows "Setup was successful".

### The decisive box

"Add python.exe to PATH" is **not ticked by default**, and it sits in small print at the very bottom. Without it, Windows will answer "python is not recognized" — the number-one failure.

**Forgot it?** Run the installer again, choose "Modify", tick it.

# macOS and Linux: one command

## macOS

**with Homebrew**

```
brew install python@3.12
```

Homebrew is the most widely used package manager on macOS. If it isn't installed: `brew.sh`. Otherwise, `python.org/downloads` also provides a macOS installer.

## Linux — Debian, Ubuntu

**with apt**

```
sudo apt install python3.12 python3.12-venv
```

The second package, `python3.12-venv`, is required: without it, creating an isolated environment fails.

On both systems, the command is called `python3`, not `python`. The "3" is part of the name.

*Neither macOS nor Linux asks about PATH: the install handles it.*

# Check that Python answers

WHAT YOU TYPE

```
> python --version
```

WHAT YOU SEE

```
Python 3.12.0
```

*Output recorded on a real machine on 18/10/2026*

**Close the terminal, open a fresh one**

Not superstition: a terminal that was already open doesn't know about software installed since.

**A version number appears: you're done**

3.12.0 or 3.12.7 — the digit after the second dot doesn't matter.

**3.10, 3.11, 3.12 and 3.13 all work**

On macOS and Linux, type `python3 --version`.

**Several Pythons on the machine?**

On Windows, `py -0p` lists them all, with their paths.

# Install JupyterLab — one command

```
python -m pip install jupyterlab
```

On macOS and Linux: `python3 -m pip install jupyterlab`

## Why `python -m pip` and not `pip`

A machine can carry several Pythons. Writing `python -m pip` installs into **the one that just answered in step 3**. Plain `pip` sometimes targets another.

## What you'll see

A long list of "Collecting…" lines, then "Successfully installed…". That's normal: JupyterLab pulls in about thirty dependencies.

**2–3**
minutes

Check: `jupyter --version` should print a block of versions, including "jupyterlab".

# Launch JupyterLab and open a notebook

**1** **Go to the project folder**

```
cd my-project
```

**2** **Launch JupyterLab**

```
jupyter lab
```

**3** **Your browser opens by itself**

address like `localhost:8888`

**4** **Open the notebook you want**

double-click it in the left-hand panel

**5** **Run a cell**

**Shift + Enter**, then the next one, top to bottom

To stop JupyterLab: go back to the terminal and press **Ctrl + C**. Closing the browser tab does not stop it.
The terminal must **stay open** the whole time you work: it is what runs JupyterLab.

# Five problems, and how to fix them

| WHAT YOU SEE | WHY | WHAT YOU DO |
|---|---|---|
| **"python is not recognized…"** | The "Add python.exe to PATH" box wasn't ticked | Re-run the installer → Modify → tick the box |
| **The Microsoft Store opens** | Windows ships a fake python.exe that points to the Store | Install Python from python.org, then reopen the terminal |
| **The version shown is 3.9 or lower** | An older Python answers first | Install a recent version; on Windows, `py -0p` lists them |
| **"pip is not recognized"** | pip exists but isn't registered with the system | Type `python -m pip …` instead of `pip …` |
| **Nothing changes after installing** | The open terminal doesn't know about newly installed software | Close it and open a fresh one |



Four of these five problems share one root: **the system doesn't know where to find Python**. None comes from a failed install — which is why reinstalling doesn't fix them.

*All were met on real machines; none is theoretical.*

YOU'RE THERE

# What you have, and what comes next

AT THIS POINT

- Python answers `python --version`
- JupyterLab opens in your browser
- You know how to run a cell: **Shift + Enter**

*The collection notebooks run on your own machine · the analysis notebook runs anywhere*